

(1)

(ii)

- 1) Decision tree construction using Entropy & Information Gain
 Given the dataset, construct a decision tree using ID3 Algorithm. calculate entropy and Information gain for each attribute.

Day	outlook	Temperature	Humidity	wind	Play Tennis
D1	Sunny	Hot	High	weak	No
D2	Sunny	Hot	High	strong	No
D3	Overcast	Hot	High	weak	Yes
D4	Rain	Mild	High	weak	Yes
D5	Rain	Cool	Normal	weak	Yes
D6	Rain	Cool	Normal	strong	No
D7	Overcast	Cool	Normal	strong	Yes
D8	Sunny	Mild	High	weak	No
D9	Sunny	Cool	Normal	weak	Yes
D10	Rain	Mild	Normal	weak	Yes
D11	Sunny	Mild	Normal	strong	Yes
D12	Overcast	Mild	High	strong	Yes
D13	Overcast	Hot	Normal	weak	Yes
D14	Rain	Mild	High	strong	No

Ans:

① calculate Total Entropy:

Total Instances = 14, Yes = 9, No = 5

$$\text{Entropy} = -\sum p_i \times \log_2(p(i))$$

$$= -\left(\frac{9}{14}\right) \log_2\left(\frac{9}{14}\right) - \left(\frac{5}{14}\right) \log_2\left(\frac{5}{14}\right)$$

$$= -0.643(-0.637) - (0.357)(-1.485)$$

$$= 0.939$$

② For Outlook,

outlook = sunny {1, 2, 8, 9, 11}, Yes = 3, No = 2

$$\text{Entropy (sunny)} = -\frac{3}{5} \log_2 \left(\frac{3}{5} \right) - \left(\frac{2}{5} \right) \log_2 \left(\frac{2}{5} \right)$$

$$= -(0.4) (-1.322) - (0.6) (-0.736)$$

$$= 0.971$$

outlook = overcast = {3, 7, 12, 13}, Yes = 4, No = 0

$$\text{Entropy (outlook)} = -\frac{4}{4} \log_2 \left(\frac{4}{4} \right) = -1(0) = 0$$

outlook = Rain = {4, 5, 6, 10, 14}, Yes = 3, No = 2

$$\text{Entropy} = \left(-\frac{3}{5} \right) \log_2 \left(\frac{3}{5} \right) - \left(\frac{2}{5} \right) \log_2 \left(\frac{2}{5} \right) \Rightarrow (-0.4) (-1.322) - (0.6) (-0.736)$$

$$= 0.971$$

weighted entropy,

$$\text{Entropy after (outlook)} = \frac{5}{14} (0.971) + \frac{4}{14} (0) + \frac{5}{14} (0.971)$$

$$= 0.693$$

$$\text{Information Gain} = 0.939 - 0.693 = 0.246$$

③ For Temperature,

Temperature = Hot {1, 2, 3, 13}, Yes = 2, No = 2

$$\text{Entropy (Hot)} = -\frac{2}{4} \log_2 \left(\frac{2}{4} \right) - \left(\frac{2}{4} \right) \log_2 \left(\frac{2}{4} \right) = 1(0.5) + 1(0.5) = 1$$

Temperature = mild {4, 8, 10, 11, 12}, Yes = 4, No = 2

$$\text{Entropy (mild)} = -\frac{4}{6} \log_2 \left(\frac{4}{6} \right) - \left(\frac{2}{6} \right) \log_2 \left(\frac{2}{6} \right) \Rightarrow (0.67) (-0.584) - (0.33) (-1.584)$$

$$= 0.918$$

Temperature = cold {5, 6, 7, 9} $\Rightarrow$ Yes = 3, No = 1

$$\text{Entropy (cool)} = -\frac{3}{4} \log_2 \left(\frac{3}{4} \right) - \left(\frac{1}{4} \right) \log_2 \left(\frac{1}{4} \right) \Rightarrow (0.75) (-0.415) - (0.25) (-2)$$

$$= 0.811$$

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weighted entropy,

$$\text{Entropy after (Temperature)} = \frac{4}{14}(1) + \frac{6}{14}(0.918) + \frac{4}{14}(0.811)$$

$$= 0.910$$

$$\text{Information Gain} = 0.939 - 0.910 = 0.029$$

④ For Humidity

Humidity = High {1, 2, 3, 4, 8, 12, 14}, Yes = 3, No = 4

$$\text{Entropy (High)} = -\frac{3}{7} \log_2 \left(\frac{3}{7} \right) - \left(\frac{4}{7} \right) \log_2 \left(\frac{4}{7} \right)$$

$$= (-0.428)(-1.223) - (0.572)(-0.807) = 0.985$$

Entropy (Normal) = {5, 6, 7, 9, 10, 11, 13}, Yes = 6, No = 1

$$\text{Entropy (Normal)} = -\frac{6}{7} \log_2 \left(\frac{6}{7} \right) - \left(\frac{1}{7} \right) \log_2 \left(\frac{1}{7} \right)$$

$$= -(0.857)(-0.223) - (0.143)(-2.807) = 0.592$$

weighted entropy,

$$\text{Entropy after (Humidity)} = \left(\frac{7}{14} \right) (0.985) + \frac{7}{14} (0.592) = 0.789$$

$$\text{Information Gain} = 0.939 - 0.789 = 0.151$$

⑤ For wind,

wind = weak {1, 3, 4, 5, 8, 9, 10, 13}, Yes = 6, No = 2

$$\text{Entropy (weak)} = -\frac{6}{8} \log_2 \left(\frac{6}{8} \right) - \left(\frac{2}{8} \right) \log_2 \left(\frac{2}{8} \right)$$

$$= (-0.75)(-0.415) - (0.25)(-2) = 0.811$$

wind = strong = {2, 6, 7, 11, 12, 14}, Yes = 3, No = 3

$$\text{Entropy (strong)} = -\frac{3}{6} \log_2 \left(\frac{3}{6} \right) - \left(\frac{3}{6} \right) \log_2 \left(\frac{3}{6} \right) = (-0.5) - 0.5 = 1$$

weighted entropy,

$$\text{Entropy after (wind)} = \frac{8}{14}(0.811) + \frac{6}{14}(1) = 0.892$$

$$\text{Information Gain} = 0.939 - 0.892 = 0.048$$

→ outlook has 0.246 (High) ⇒ ROOT NODE

→ temperature has 0.029

→ humidity has 0.151

→ wind has 0.048

For Sunny,

Day	Temp	Humidity	wind	Play Tennis
1	Hot	High	weak	No
2	Hot	High	strong	No
8	mild	High	weak	No
9	cool	Normal	weak	Yes
11	Mild	Normal	strong	Yes

① Total Entropy

$$\text{Total} = 5, \text{Yes} = 2, \text{No} = 3$$

$$\text{Entropy} = -\sum p_i \times \log_2(p_i)$$

$$= -\frac{2}{5} \log_2\left(\frac{2}{5}\right) - \left(\frac{3}{5}\right) \log_2\left(\frac{3}{5}\right)$$

$$= 0.971$$

② For Temperature,

$$\text{Temperature} = \text{Hot} \{1, 2\}, \text{Yes} = 0, \text{No} = 2$$

$$\text{Entropy (Hot)} = -\frac{2}{2} \log_2\left(\frac{2}{2}\right) = (-1)(0) = 0$$

$$\text{Temperature} = \text{mild} \{8, 11\}, \text{Yes} = 1, \text{No} = 1$$

$$\text{Entropy (mild)} = -\frac{1}{2} \log_2\left(\frac{1}{2}\right) - \left(\frac{1}{2}\right) \log_2\left(\frac{1}{2}\right) = (-0.5) + (-0.5) = 1$$

$$\text{Temperature} = \text{cool} \{9\}, \text{Yes} = 1, \text{No} = 0$$

$$\text{Entropy (cool)} = -\frac{1}{1} \log_2\left(\frac{1}{1}\right) = (-1)(0) = 0$$

weighted Entropy,

$$\text{Entropy after (Temperature)} = \frac{2}{5}(0) + \frac{2}{5}(1) + \frac{1}{5}(0) = 0.4$$

$$\text{Information Gain} = 0.971 - 0.4 = 0.571$$

③ For Humidity,

$$\text{Humidity} = \text{High} \{1, 2, 3\}, \text{No} = 3, \text{Yes} = 0$$

$$\text{Entropy} = -\frac{3}{3} \log_2\left(\frac{3}{3}\right) = (-1)(0) = 0$$

$$\text{Humidity} = \text{Normal} \{9, 11\}, \text{Yes} = 2, \text{No} = 0$$

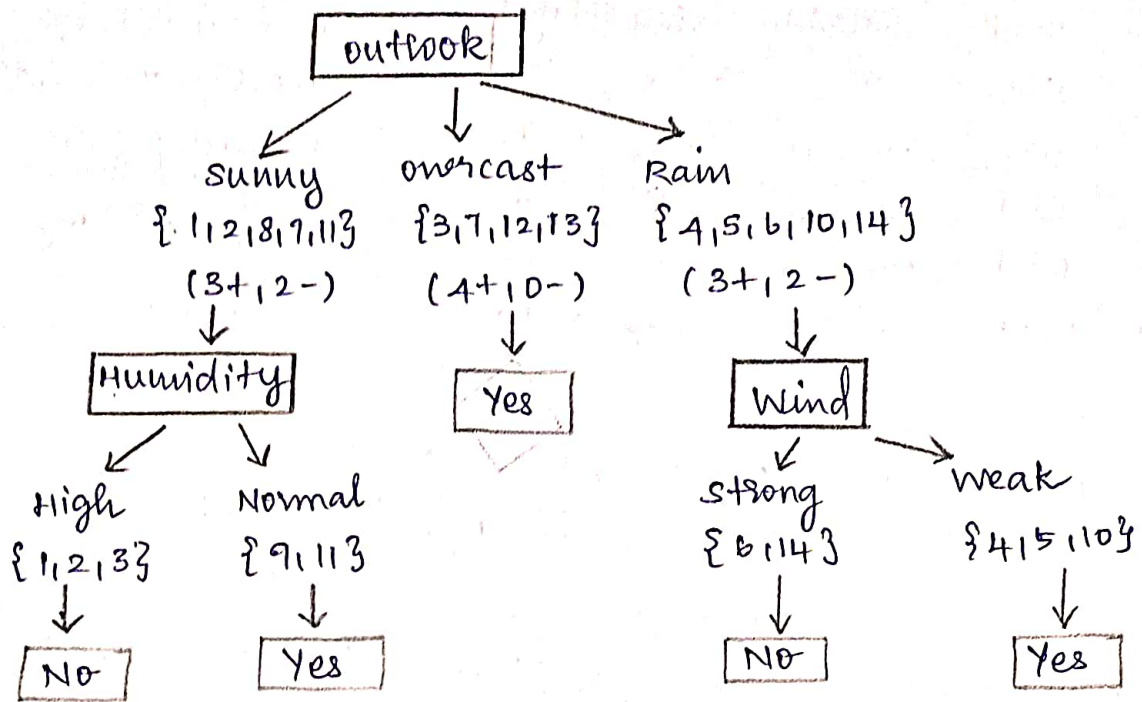
$$\text{Entropy} = \left(-\frac{2}{2}\right) \log_2\left(\frac{2}{2}\right) = (-2)(0) = 0$$

weighted Entropy,

$$\text{Entropy after (Humidity)} = \frac{3}{5}(0) + \frac{2}{5}(0) = 0$$

$$\text{Information Gain} = 0.971$$

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④ For wind,

wind = weak {1, 8, 9}, Yes = 1, No = 2

$$\text{Entropy(weak)} = -\frac{1}{3} \log_2 \left(\frac{1}{3} \right) - \left(\frac{2}{3} \right) \log_2 \left(\frac{2}{3} \right) = (-0.33)(-1.584) - (0.67)(0.584) = 0.9183$$

Entropy $\Rightarrow$ wind = strong {2, 11}, Yes = 1, No = 1

$$\Rightarrow -\frac{1}{2} \log_2 \left(\frac{1}{2} \right) - \left(\frac{1}{2} \right) \log_2 \left(\frac{1}{2} \right) = 0.5 + 0.5 = 1$$

weighted entropy,

$$\text{Entropy after (wind)} = \frac{3}{5} (0.9183) + \frac{2}{5} (1) = 0.950$$

$$\text{Information Gain} = 0.971 - 0.950 = 0.02$$

$\rightarrow$ Temperature has 0.571

$\rightarrow$ Humidity has 0.971 (High)

$\rightarrow$ wind has 0.02

Day	Temp	Humidity	wind	Play Tennis
4	Mild	High	weak	Yes
5	cool	Normal	strong	Yes
6	cool	Normal	weak	No
10	mild	Normal	weak	Yes
14	mild	High	strong	No

① Total Entropy

Total = 5, Yes = 3, No = 2

$$\text{Entropy} = \left(-\frac{2}{5}\right) \log_2 \left(\frac{2}{5}\right) - \left(\frac{3}{5}\right) \log_2 \left(\frac{3}{5}\right)$$

$$= 0.971$$

② Temperature,

Temperature = Mild {4, 10, 14}, Yes = 2, No = 1

$$\text{Entropy (Mild)} = -\frac{2}{3} \log_2 \left(\frac{2}{3}\right) - \left(\frac{1}{3}\right) \log_2 \left(\frac{1}{3}\right) = (0.67)(0.584) + (0.33)(1.584) = 0.918$$

Temperature = cool {5, 6}, Yes = 1, No = 1

$$\text{Entropy (cool)} = \left(-\frac{1}{2}\right) \log_2 \left(\frac{1}{2}\right) - \left(\frac{1}{2}\right) \log_2 \left(\frac{1}{2}\right) = (0.5) + (0.5) = 1$$

weighted entropy,

$$\text{Entropy after (Temperature)} = \frac{2}{5}(1) + \frac{3}{5}(0.918) = 0.950$$

$$\text{Information Gain} = 0.971 - 0.950 = 0.02$$

③ Humidity,

Humidity = High {4, 14}, Yes = 1, No = 1

$$\text{Entropy (High)} = -\frac{1}{2} \log_2 \left(\frac{1}{2}\right) - \left(\frac{1}{2}\right) \log_2 \left(\frac{1}{2}\right) = (-0.5) - (-0.5) = 1$$

Humidity = Normal = {5, 6, 10}, Yes = 2, No = 1

$$\text{Entropy (Normal)} = -\frac{2}{3} \log_2 \left(\frac{2}{3}\right) - \left(\frac{1}{3}\right) \log_2 \left(\frac{1}{3}\right) = 0.918$$

weighted entropy,

$$\text{Entropy after (Humidity)} = \frac{2}{5}(1) + \frac{3}{5}(0.918) = 0.950$$

$$\text{Information Gain} = 0.971 - 0.950 = 0.02$$

④ wind,

wind = weak = {4, 5, 10}, Yes = 3, No = 2, Entropy = $\left(-\frac{3}{5}\right) \log_2 \left(\frac{3}{5}\right) = 0$

wind = strong = {6, 14}, Yes = 0, No = 2, Entropy = $\left(-\frac{2}{2}\right) \log_2 \left(\frac{2}{2}\right) = 0$

weighted entropy,

$$\text{Entropy after (Wind)} = \frac{3}{5}(0) + \frac{2}{5}(0) = 0$$

$$\text{Information Gain} = 0.971$$

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(ii)

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use the dataset loan approval, apply ID3 algorithm. Calculate Entropy and Information Gain for each attribute & identify Root Node

ID	Credit score	Income	Employment	Loan Approved
1	Good	High	Yes	Yes
2	Good	High	No	Yes
3	Fair	High	Yes	Yes
4	Poor	medium	Yes	No
5	Poor	Low	Yes	No
6	Poor	Low	No	No
7	Fair	Low	No	Yes
8	Good	medium	Yes	Yes
9	Good	Low	Yes	No
10	Poor	medium	No	No
11	Good	medium	No	Yes
12	Fair	medium	Yes	Yes
13	Fair	High	No	Yes
14	Poor	medium	Yes	No

① Total Entropy,

Total instances = 14, Yes = 8, No = 6

$$\text{Entropy} = -\frac{8}{14} \log_2 \left(\frac{8}{14} \right) - \left(\frac{6}{14} \right) \log_2 \left(\frac{6}{14} \right)$$

$$= -(0.571)(-0.807) - (0.429)(-1.223) = 0.985$$

② For Credit score,

Credit score = Good = {1, 2, 8, 9, 11}, Yes = 4, No = 1

$$\text{Entropy (Good)} = -\frac{4}{5} \log_2 \left(\frac{4}{5} \right) - \left(\frac{1}{5} \right) \log_2 \left(\frac{1}{5} \right) = -(0.8)(-0.322) - (0.2)(-2.321)$$

$$= 0.722$$

Credit score = Fair = {3, 7, 12, 13}, Yes = 4, No = 0

$$\text{Entropy (Fair)} = -\frac{4}{4} \log_2 \left(\frac{4}{4} \right) = (-1)(0) = 0$$

Credit score = Poor = {4, 5, 6, 10, 14}, Yes = 0, No = 5

$$\text{Entropy (Poor)} = -\frac{5}{5} \log_2 \left(\frac{5}{5} \right) = (-1)(0) = 0$$

$$\text{weighted average} \Rightarrow \frac{5}{14}(0.722) + \frac{4}{14}(0) + \frac{5}{14}(0) = 0.258$$

$$\text{Information Gain} = 0.985 - 0.258 = 0.727$$

③ For Income,

Income = High = {1, 2, 3, 13}, Yes = 4, No = 0, Entropy = $(-\frac{4}{4}) \log_2(\frac{4}{4}) = 0$

Income = medium = {4, 8, 10, 11, 12}, Yes = 3, No = 3

$$\text{Entropy (medium)} = -\frac{3}{6} \log_2(\frac{3}{6}) - (\frac{3}{6}) \log_2(\frac{3}{6}) = 1$$

Income = Low = {5, 6, 7, 9}, Yes = 1, No = 3

$$\text{Entropy (Low)} = (-\frac{1}{4}) \log_2(\frac{1}{4}) - (\frac{3}{4}) \log_2(\frac{3}{4}) = 0.811$$

$$\text{weighted Entropy} = \frac{4}{14} (0) + \frac{6}{14} (1) + \frac{4}{14} (0.811) = 0.660$$

$$\text{Information Gain} = 0.985 - 0.660 = 0.325$$

④ For Employment,

Employment = Yes {1, 3, 4, 5, 8, 9, 12, 14}, Yes = 4, No = 4

$$\text{Entropy (yes)} = -\frac{4}{8} \log_2(\frac{4}{8}) - (\frac{4}{8}) \log_2(\frac{4}{8}) = 0.5 + 0.5 = 1$$

Employment = No {2, 6, 7, 10, 11, 13}, Yes = 4, No = 2

$$\text{Entropy (No)} = -\frac{4}{6} \log_2(\frac{4}{6}) - \frac{2}{6} \log_2(\frac{2}{6}) = 0.918$$

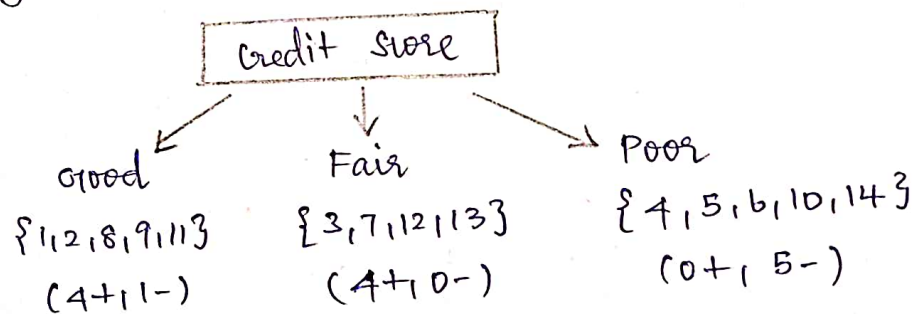
$$\text{weighted Entropy} = \frac{8}{14} (1) + \frac{6}{14} (0.918) = 0.964$$

$$\text{Information Gain} = 0.985 - 0.964 = 0.021$$

→ Credit score has 0.727 (High) => ROOT NODE

→ Income has 0.325

→ Employment has 0.021



(5)

- (ii) 3) A Supermarket has following transaction database. Apply apriori algorithm with min-support = 50% & min-confidence = 70%. Find all frequent itemsets and strong association rules. Calculate lift for each rule.

TID	Items Purchased
T1	Bread, milk, Eggs
T2	Bread, Butter
T3	milk, Butter, Cheese
T4	Bread, milk, Butter
T5	Bread, milk, Eggs, Butter
T6	milk, Eggs
T7	Bread, Eggs
T8	milk, Butter.

Ans: ① Create a table containing support count for each item

min-support = 50%

min-confidence = 70%

Total Transactions = 8

$$\text{Min support count} = 50\% \times 8 = \frac{150}{100} \times 8 = 4$$

Frequent 1-Itemset,

Itemset	support count	support
Bread	5 (≥ 4)	$5/8 = 62.5\%$
Milk	6 (≥ 4)	$6/8 = 75\%$
Butter	5 (≥ 4)	$5/8 = 62.5\%$
Eggs	4 (≥ 4)	$4/8 = 50\%$
Cheese	1 ($\times$)	$1/8 = 12.5\%$

$$\text{support} = \frac{\text{No. of Transactions containing}}{\text{Total Transactions}}$$

Frequent 1-Itemset $\Rightarrow \{ \text{Bread} \}, \{ \text{milk} \}, \{ \text{Butter} \}, \{ \text{Eggs} \}$

Frequent 2-Itemsets,

Itemset	support-count	support
Bread, milk	3 (x)	$3/8 = 37.5\%$
Bread, Butter	3 (x)	$3/8 = 37.5\%$
Bread, Eggs	3 (x)	$3/8 = 37.5\%$
milk, Butter	4 (≥ 4)	$4/8 = 50\%$
milk, Eggs	3 (x)	$3/8 = 37.5\%$
Butter, Eggs	1 (x)	$1/8 = 12.5\%$

Frequent 2-Itemsets $\Rightarrow \{ \text{milk, Butter} \}$

Frequent 3-Itemsets,

Frequent 3-Itemsets can't be formed, only one frequent 2-Itemsets exists.

② Generate Association Rules,

min-confidence = 70%

Rule 1: milk $\rightarrow$ Butter

$$\text{confidence}(A \rightarrow B) = \frac{\text{support_count}(A \cup B)}{\text{support_count}(A)}$$

$$\text{confidence}(\text{milk} \rightarrow \text{Butter}) = \frac{\text{support_count}(\text{milk} \cup \text{Butter})}{\text{support_count}(\text{milk})} = \frac{4}{6} = 66.67\%$$

$66.67\% < 70\%$ (Not strong)

Rule 2: Butter $\rightarrow$ milk

$$\text{confidence}(\text{Butter} \rightarrow \text{milk}) = \frac{\text{support_count}(\text{Butter} \cup \text{milk})}{\text{support_count}(\text{Butter})} = \frac{4}{5} = 80\%$$

$80\% > 70\%$ (strong)

③ Calculate Lift

$$\text{Lift} = \frac{\text{confidence}(A \rightarrow B)}{\text{support}(B)}$$

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For Buttes $\rightarrow$ Milk,

$$\text{Lift} = \frac{\text{confidence (Buttes} \rightarrow \text{Milk)}}{\text{Support (Milk)}} = \frac{80\%}{75\%} = \frac{0.80}{0.75} = 1.07$$

Strong Association Rule is Buttes $\rightarrow$ Milk,

Support = 50%

confidence = 80%

Lift = 1.07

- 4) Apply the Apriori Algorithm on the following transaction with min-support = 40% and min-confidence = 60%. Generate all association rules and support, confidence and Lift

TID	Books Purchased
T1	Python, ML, Statistics
T2	Python, Data science
T3	ML, Statistics, Data science
T4	Python, ML
T5	Python, ML, Data science

Ans: ① Create a table containing support count for each item

min-support = 40%

min-confidence = 60%

Total Transactions = 5

$$\text{min support count} = 40\% \times 5 = \frac{40}{100} \times 5 = 2$$

Frequent 1-Itemset,

Itemset	Support-count	Support
Python	4 ($>= 2$)	$4/5 = 80\%$
ML	4 ($>= 2$)	$4/5 = 80\%$
Statistics	2 ($>= 2$)	$2/5 = 40\%$
Data science	3 ($>= 2$)	$3/5 = 60\%$

Frequent 1-Itemset $\Rightarrow \{Python\}, \{ML\}, \{Statistics\}, \{Data\ science\}$

Frequent 2-Itemset,

Itemset	Support-count	Support
Python, ML	3 (≥ 2)	$3/5 = 60\%$
Python, Statistics	1 (x)	$1/5 = 20\%$
Python, Data science	2 (≥ 2)	$2/5 = 40\%$
ML, Statistics	2 (≥ 2)	$2/5 = 40\%$
ML, Data science	2 (≥ 2)	$2/5 = 40\%$
Statistics, Data science	1 (x)	$1/5 = 20\%$

Frequent 2-Itemsets $\Rightarrow \{Python, ML\}, \{Python, Data\ science\},$
 $\{ML, Statistics\}, \{ML, Data\ science\}$

Frequent 3-Itemset,

Python, ML, Data science, Statistics

Itemset	Support-count	Support
Python, ML, Data science	1 (x)	$1/5 = 20\%$
Python, ML, Statistics	1 (x)	$1/5 = 20\%$
ML, Data science, Statistics	1 (x)	$1/5 = 20\%$

All Itemsets have frequent $[< 2]$, so, stop

② Generate Association Rules, min-confidence = 60%

Rule 1: $Python \rightarrow ML$

$$\text{Confidence}(Python \rightarrow ML) = \frac{\text{Confidence}(Python \cup ML)}{\text{Support-count}(Python)} = \frac{3}{4} = 75\% > 60\% \quad (\text{strong})$$

Rule 2: $ML \rightarrow Python$

$$\text{Confidence}(ML \rightarrow Python) = \frac{\text{Confidence}(ML \cup Python)}{\text{Support-count}(ML)} = \frac{3}{4} = 75\% > 60\% \quad (\text{strong})$$

Rule 3: Python $\rightarrow$ Data science

$$\text{confidence (Python} \rightarrow \text{Data science)} = \frac{\text{confidence (Python} \cup \text{Data science)}}{\text{support-count (Python)}} \\ = \frac{2}{4} = 50\% < 60\% \text{ (Not strong)}$$

Rule 4: Data science $\rightarrow$ Python

$$\text{confidence (Data science} \rightarrow \text{Python)} = \frac{\text{confidence (Data science} \cup \text{Python)}}{\text{support-count (Data science)}} = \frac{2}{3} = 66.7\% \\ 66.7\% > 60\% \text{ (strong)}$$

Rule 5: ML $\rightarrow$ Statistics

$$\text{confidence (ML} \rightarrow \text{Statistics)} = \frac{\text{confidence (ML} \rightarrow \text{Statistics)}}{\text{support-count (ML)}} = \frac{2}{4} = 50\% < 60\% \text{ (Not strong)}$$

Rule 6: Statistics $\rightarrow$ ML

$$\text{confidence (Statistics} \rightarrow \text{ML)} = \frac{\text{confidence (Statistics} \rightarrow \text{ML)}}{\text{support-count (Statistics)}} = \frac{2}{2} = 100\% \text{ (strong)}$$

Rule 7: ML $\rightarrow$ Data science

$$\text{confidence (ML} \rightarrow \text{Data science)} = \frac{\text{confidence (ML} \cup \text{Data science)}}{\text{support-count (ML)}} \\ = \frac{2}{4} = 50\% < 60\% \text{ (Not strong)}$$

Rule 8: Data science $\rightarrow$ ML

$$\text{confidence (Data science} \rightarrow \text{ML)} = \frac{\text{confidence (Data science} \cup \text{ML)}}{\text{support-count (Data science)}} \\ = \frac{2}{3} = 66.7\% > 60\% \text{ (strong)}$$

③ Calculate Lift

$$\text{Lift} = \frac{\text{confidence (A} \rightarrow \text{B)}}{\text{support (B)}}$$

1: Python $\rightarrow$ ML

$$\text{Lift} = \frac{\text{Confidence}(\text{Python} \rightarrow \text{ML})}{\text{Support}(\text{ML})} = \frac{75\%}{80\%} = \frac{0.75}{0.80} = 0.93$$

2: ML $\rightarrow$ Python

$$\text{Lift} = \frac{\text{Confidence}(\text{ML} \rightarrow \text{Python})}{\text{Support}(\text{Python})} = \frac{75\%}{80\%} = \frac{0.75}{0.80} = 0.93$$

3: DataScience $\rightarrow$ Python

$$\text{Lift} = \frac{\text{Confidence}(\text{DataScience} \rightarrow \text{Python})}{\text{Support}(\text{Python})} = \frac{66.7\%}{80\%} = \frac{0.667}{0.80} = 0.83$$

4: Statistics $\rightarrow$ ML

$$\text{Lift} = \frac{\text{Confidence}(\text{Statistics} \rightarrow \text{ML})}{\text{Support}(\text{ML})} = \frac{100\%}{80\%} = \frac{1}{0.80} = 1.25$$

5: DataScience $\rightarrow$ ML

$$\text{Lift} = \frac{\text{Confidence}(\text{DataScience} \rightarrow \text{ML})}{\text{Support}(\text{ML})} = \frac{66.7\%}{80\%} = \frac{0.667}{0.80} = 0.83$$

Rule	Lift	Interpretation
Python $\rightarrow$ ML	0.93	slight -ve association
ML $\rightarrow$ Python	0.93	slight a -ve association
DataScience $\rightarrow$ Python	0.83	-ve association
Statistics $\rightarrow$ ML	1.25	+ve association
DataScience $\rightarrow$ ML	0.83	-ve association

* Statistics $\rightarrow$ ML has Positive Lift value

* All others have weak or Negative Lift value

(iii)

- 5) The following matrix shows ratings (1-5) by users. Generate recommendations for 5 users using user-based collaborative filtering. Use Pearson correlation to find similar users. Predict rating for movie 4.

User	Book 1	Book 2	Book 3	Book 4	Movie 5
User 1	5	3	4	4	2
User 2	3	1	2	3	3
User 3	4	3	4	3	5
User 4	3	3	1	5	4
User 5	1	5	5	?	2

Ans:

- ① From the matrix,

$$\text{User 1: } [5, 3, 4, 2]$$

$$\text{User 2: } [3, 1, 2, 3]$$

$$\text{User 3: } [4, 3, 4, 5]$$

$$\text{User 4: } [3, 3, 1, 4]$$

$$\text{User 5: } [1, 5, 5, 2]$$

- ② Calculate similarity using cosine similarity

$$\text{cosine similarity} = \frac{\sum_{i=1}^n A_i B_i}{\sqrt{\sum_{i=1}^n A_i^2} \times \sqrt{\sum_{i=1}^n B_i^2}}$$

Similarity between user 5 and user 1:

common ratings: $[1, 5, 5, 2]$ and $[5, 3, 4, 2]$

$$\text{Numerator} = (1 \times 5) + (5 \times 3) + (5 \times 4) + (2 \times 2) = 5 + 15 + 20 + 4 = 44$$

$$\text{Alice's magnitude} = \sqrt{1^2 + 5^2 + 5^2 + 2^2} = \sqrt{55} = 7.416$$

$$\text{user 1's magnitude} = \sqrt{5^2 + 3^2 + 4^2 + 2^2} = \sqrt{54} = 7.348$$

$$\text{Similarity (user 5, user 1)} = \frac{44}{(7.416 \times 7.348)} = \frac{44}{54.49} = 0.807$$

Similarity between user 5 and user 2:

common ratings: $[1, 5, 5, 2]$ and $[3, 1, 2, 3]$

$$\text{Numerator} = (1 \times 3) + (5 \times 1) + (5 \times 2) + (2 \times 3) = 24$$

$$\text{user 5's magnitude} = \sqrt{1^2 + 5^2 + 5^2 + 2^2} = \sqrt{55} = 7.416$$

$$\text{user 2's magnitude} = \sqrt{3^2 + 1^2 + 2^2 + 3^2} = \sqrt{9 + 1 + 4 + 9} = \sqrt{23} =$$

$$\text{Similarity (user 5, user 2)} = 24 / (7.416 \times 2.189)$$

$$= 24 / 16.23 = 1.478$$

user 5 and user 3 :

$$\text{user 5} \Rightarrow [1, 5, 5, 2] \quad , \quad \text{user 3} = [4, 3, 4, 5]$$

$$\text{Numerator} = (1 \times 4) + (5 \times 3) + (5 \times 4) + (2 \times 5) = 4 + 15 + 20 + 10 = 49$$

$$\text{user 5's magnitude} = \sqrt{1^2 + 5^2 + 5^2 + 2^2} = \sqrt{55} = 7.416$$

$$\text{user 3's magnitude} = \sqrt{4^2 + 3^2 + 4^2 + 5^2} = \sqrt{16 + 16 + 9 + 25} = \sqrt{66} = 8.124$$

$$\text{Similarity (user 5, user 3)} = 49 / (7.416 \times 8.124) = 49 / 60.247 = 0.813$$

user 5 and user 4 :

$$\text{user 5} \Rightarrow [1, 5, 5, 2] \quad , \quad \text{user 4} = [3, 3, 1, 4]$$

$$\text{Numerator} = (1 \times 3) + (5 \times 3) + (5 \times 1) + (2 \times 4) = 3 + 15 + 5 + 8 = 31$$

$$\text{user 5's magnitude} = \sqrt{1^2 + 5^2 + 5^2 + 2^2} = \sqrt{55} = 7.416$$

$$\text{user 4's magnitude} = \sqrt{3^2 + 3^2 + 1^2 + 4^2} = \sqrt{9 + 9 + 1 + 16} = \sqrt{35} = 5.916$$

$$\text{Similarity (user 5, user 4)} = 31 / (7.416 \times 5.916) = 31 / 43.873 = 0.706$$

user	similarity with user 5
user 1	0.807
user 2	1.478 (High)
user 3	0.813
user 4	0.706

③ Predict ~~the~~ user 5's Rating for movie 4

$$\text{Predicted Rating} = \frac{\sum (\text{similarity}_i \times \text{Rating}_i)}{\sum \text{similarity}_i}$$

$$\text{user 1 : } 4$$

$$\text{user 2 : } 3$$

$$\text{user 3 : } 3$$

$$\text{user 4 : } 5$$

$$\text{Predicted Rating} = \frac{(0.807 \times 4) + (1.478 \times 3) + (0.813 \times 3) + (0.706 \times 5)}{0.807 + 1.478 + 0.813 + 0.706}$$

$$= \frac{13.631}{3.804} = 3.583 \approx 4$$

Predicted Rating for movie 4 by user 5 is 4 (3.583)

- 6) using the book ratings, apply user-based CF with cosine similarity to predict what rating user A given to Book 4. user top 2 similar users.

user	Book 1	Book 2	Book 3	Book 4
user A	4	0	5	?
user B	5	4	4	2
user C	0	0	0	5
user D	2	4	0	4
user E	3	3	1	3

Ans: ① From the matrix,

user A: [4, 5]

user B: [5, 4]

user C: [0, 0]

user D: [2, 0]

user E: [3, 4]

For user A, Book 2 is 0 & Book 4 is ?
If we include it, similarity becomes incorrect, so ignore it

- ② Calculate similarity using cosine similarity

Similarity between user A and user B,
common ratings: [4, 5] and [5, 4]

$$\text{Numerator} = (4 \times 5) + (5 \times 4) = 40$$

$$\text{user A's magnitude} = \sqrt{4^2 + 5^2} = \sqrt{16 + 25} = \sqrt{41} = 6.403$$

$$\text{user B's magnitude} = \sqrt{5^2 + 4^2} = \sqrt{41} = 6.403$$

$$\text{Similarity (user A, user B)} = 40 / (6.403 \times 6.403) = 40 / 40.99 = 0.975$$

Similarity between user A and user C,

common ratings: [4, 5] and [0, 0]

$$\text{Numerator} = (4 \times 0) + (5 \times 0) = 0$$

$$\text{user A's magnitude} = \sqrt{41} = 6.403, \text{ user C's magnitude} = 0$$

$$\text{Similarity (user A, user C)} = 0 / (6.403 \times 0) = 0$$

Similarity between user A and user D

Common Ratings: [4, 5] and [2, 0], numerator = ~~4~~ $4 \times 2 = 8$

$$\text{user A's magnitude} = \sqrt{4^2} = \sqrt{16} = 4$$

$$\text{user D's magnitude} = \sqrt{2^2} = \sqrt{4} = 2$$

$$\begin{bmatrix} 4, 5 \\ 2, 0 \end{bmatrix}$$

$$\text{Similarity (user A, user D)} = 8 / 2(4) = 8 / 12.80 = 1$$

similarity between user A and user E
common ratings: [4, 5] and [3, 4]

$$\text{user A's magnitude} = \sqrt{41} = 6.403$$

$$\text{user E's magnitude} = \sqrt{9+16} = \sqrt{25} = 5$$

$$\text{Numerator} = (4 \times 3) + (5 \times 4) = 12 + 20 = 32$$

$$\text{similarity (user A, user E)} =$$

$$\frac{32}{6.403 \times 5} = \frac{32}{32.015} = 0.999$$

user	similarity with user A
user B	0.975
user C	0
user D	1
user E	0.999

Top-2 Similar Users,

user D $\rightarrow 1$, user E $\rightarrow 0.999$

③ Predict user A's Rating for Book 4

user B : 2

user C : 5

user D : 4 (✓)

user E : 3 (✓)

$$\text{Predicted Rating} = \frac{(4 \times 1) + (0.999 \times 3)}{1 + 0.999} = \frac{4 + 2.997}{1.999} = 3.5 \approx 4$$

Predicted Rating for user A Rating for Book 4 is 4.

(iv)

7) using the following user-item rating matrix, apply item-based collaborative filtering with cosine similarity to predict the rating user C would give to Item 3

User	Item 1	Item 2	Item 3	Item 4	Item 5
User A	4	3	0	5	1
User B	5	0	4	4	2
User C	3	1	?	3	2
User D	0	0	4	0	0
User E	1	0	5	2	0

Ans:

Missing Ratings, user C $\rightarrow$ Item 3

① Built Item-Item Similarity matrix

$$\text{Sim}(i, j) = \frac{\sum_{u \in U_{ij}} (r_{u,i} - \bar{r}_u)(r_{u,j} - \bar{r}_u)}{\sqrt{\sum_{u \in U_{ij}} (r_{u,i} - \bar{r}_u)^2} \times \sqrt{\sum_{u \in U_{ij}} (r_{u,j} - \bar{r}_u)^2}}$$

User Averages,

$$\text{user A} = (4+3+5+1)/4 = 13/4 = 3.25$$

$$\text{user B} = (5+4+4+2)/4 = 15/4 = 3.75$$

$$\text{user C} = (3+1+3+2)/4 = 9/4 = 2.25$$

$$\text{user D} = 4/1 = 4$$

$$\text{user E} = (1+5+2)/3 = 8/3 = 2.67$$

② calculate item similarities

Similarity (Item 3, Item 1)

Common users: user B, user E

User	Item 1	Item 3	avg	$r_1 - \text{avg}$	$r_3 - \text{avg}$	Prod	$(r_1 - \text{avg})^2$	$(r_3 - \text{avg})^2$
B	5	4	3.75	1.25	0.25	0.3125	1.5625	0.0625
E	1	5	2.67	-1.67	2.33	-3.89	2.79	5.43

$$\text{Numerator} = 0.3125 - 3.89 = -3.5775$$

$$\text{Denominator} = \sqrt{(1.5625 + 2.79)} \times \sqrt{(0.0625 + 5.43)}$$

$$= \sqrt{5.4925} \times \sqrt{5.4925}$$

$$= 2.344 \times 2.086 = 4.887$$

$$\text{Sim}(\text{Item 3, Item 1}) = -3.5775 / 4.887 = -0.732$$

Similarity (Item 3, Item 4)

Common users: user B, user E

User	Item 3	Item 4	avg	$r_3 - \text{avg}$	$r_4 - \text{avg}$	Prod	$(r_3 - \text{avg})^2$	$(r_4 - \text{avg})^2$
B	4	4	3.75	0.25	0.25	0.0625	0.0625	0.0625
E	5	2	2.67	2.33	-0.67	-1.561	5.43	0.45

$$\text{Numerator} = 0.0625 - 1.561 = -1.4985$$

$$\text{Denominator} = \sqrt{0.0625 + 5.43} \times \sqrt{0.0625 + 0.45}$$

$$= \sqrt{5.4925} \times \sqrt{0.5114}$$

$$= 2.344 \times 0.715 = 1.676$$

$$\text{Sim}(\text{Item 3}, \text{Item 4}) = -1.4985 / 1.676 = -0.894$$

similarity (Item 3, Item 2) $\Rightarrow$ No valid common users

similarity (Item 3, Item 5)

common users: User B

users	Item 3	Item 5	avg	(r3-avg)	(r5-avg)	prd	(r3-avg) ²	(r5-avg) ²
B	4	2	3.75	0.25	-1.75	-0.4375	0.0625	1.5625

$$\text{Numerator} = -0.4375$$

$$\text{Denominator} = \sqrt{0.0625} \times \sqrt{1.5625} = 0.25 \times 1.25 = 0.3125$$

$$\text{Sim}(\text{Item 3}, \text{Item 5}) = -0.4375 / 0.3125 = -1.4$$

Item similarity Matrix,

	Item 3
Item 1	-0.732
Item 4	-0.894
Item 5	-1.4

③ Predict missing Ratings,

$$\hat{r}_{u,i} = \frac{\sum_{j \in N(u)} \text{sim}(i,j) \times r_{u,j}}{\sum_{j \in N(u)} |\text{sim}(i,j)|}$$

Prediction 1: user c's rating for Item 3, user c rated Item 1, 2, 4, 5

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	User c's rating
Item 1	3
Item 2	1
Item 4	3
Item 5	2

$$\hat{r}_{c13} = \frac{\text{sim}(3,1)r_{c,1} + \text{sim}(3,2)r_{c,2} + \text{sim}(3,4)r_{c,4} + \text{sim}(3,5)r_{c,5}}{|\text{sim}(3,1)| + |\text{sim}(3,2)| + |\text{sim}(3,4)| + |\text{sim}(3,5)|}$$

Since $\text{sim}(3,2)$ doesn't have common user avoid it

$$\begin{aligned}\hat{r}_{c13} &= \frac{(-0.732 \times 3) + (-1.4985 \times 3) + (-1.4 \times 2)}{0.732 + 1.4985 + 1.4} \\ &= \frac{-2.196 - 4.4955 - 2.8}{3.63} = \frac{-9.49}{3.63} = 2.6 \approx 3\end{aligned}$$

Use absolute similarities (without sign),

$$\hat{r}_{c13} = 3$$

Final Answer,

User	Item 1	Item 2	Item 3	Item 4	Item 5
A	4	3	0	5	1
B	5	0	4	4	2
C	3	1	3	3	2
D	0	0	4	0	0
E	1	0	5	2	0

- 8) Apply Item-based CF using adjusted cosine similarity on the following data. Predict the ratings of user 2 and user 5 would give to Prod 5.

User	P1	P2	P3	P4	P5
user 1	3	4	3	1	2
user 2	4	3	4	1	?
user 3	3	3	1	2	4
user 4	1	1	2	3	3
user 5	2	1	5	1	?

Ans:

Missing Ratings, user 2 $\rightarrow$ P5,
user 5 $\rightarrow$ P5

① user Averages,

$$\text{user 1} = (3+4+3+1+2)/5 = 13/5 = 2.6$$

$$\text{user 2} = (4+3+4+1)/4 = 12/4 = 3$$

$$\text{user 3} = (3+3+1+2+4)/5 = 13/5 = 2.6$$

$$\text{user 4} = (1+1+2+3+3)/5 = 10/5 = 2$$

$$\text{user 5} = (2+1+5+1)/4 = 9/4 = 2.25$$

② calculate Item similarities

similarity (P5, P1)

user	P5	P1	avg	(r5-avg)	(r1-avg)	Prod	(r5-avg) ²	(r1-avg) ²
user 1	2	3	2.6	-0.6	0.4	-0.24	0.36	0.16
user 3	4	3	2.6	1.4	0.4	0.56	1.96	0.16
user 4	3	1	2	1	-1	-1	1	1

$$\text{Numerator} = -0.24 + 0.56 - 1 = -0.68$$

$$\text{Denominator} = \sqrt{0.36 + 1.96 + 1} \times \sqrt{0.16 + 0.16 + 1}$$

$$= \sqrt{3.32} \times \sqrt{1.32}$$

$$= 1.822 \times 1.149 = 2.092$$

$$\text{Sim}(P5, P1) = -0.68 / 2.092 = -0.325$$

Similarity (P5, P2)

user	P5	P2	avg	(r5-avg)	(r2-avg)	prd	(r5-avg) ²	(r2-avg) ²
user1	2	4	2.6	-0.6	1.4	-0.84	0.36	1.96
user3	4	3	2.6	1.4	0.4	0.56	1.96	0.16
user4	3	1	2	1	-1	-1	1	1

$$\text{Numerator} = -0.84 + 0.56 - 1 = -1.28$$

$$\begin{aligned}\text{Denominator} &= \sqrt{0.36 + 1.96 + 1} + \sqrt{1.96 + 0.16 + 1} \\ &= \sqrt{3.32} \times \sqrt{3.12} \\ &= 1.822 \times 1.767 = 3.22\end{aligned}$$

$$\text{Sim}(P5, P2) = -1.28 / 3.22 = -0.398$$

Similarity (P5, P3)

user	P5	P3	avg	(r5-avg)	(r3-avg)	prd	(r5-avg) ²	(r3-avg) ²
user1	2	3	2.6	-0.6	0.4	-0.24	0.36	0.16
user3	4	1	2.6	1.4	-1.6	-2.24	1.96	2.56
user4	3	2	2	1	0	0	1	0

$$\text{Numerator} = -0.24 - 2.24 + 0 = -2.48$$

$$\begin{aligned}\text{Denominator} &= \sqrt{0.36 + 1.96 + 1} \times \sqrt{0.16 + 2.56} \\ &= \sqrt{3.32} \times \sqrt{2.72} \\ &= 1.822 \times 1.649 = 3.005\end{aligned}$$

$$\text{Sim}(P5, P3) = -2.48 / 3.005 = -0.825$$

Similarity (P5, P4)

user	P5	P4	avg	(r5-avg)	(r4-avg)	prd	(r5-avg) ²	(r4-avg) ²
user1	2	1	2.6	-0.6	-1.6	0.96	0.36	2.56
user3	4	2	2.6	1.4	-0.6	-0.84	1.96	0.36
user4	3	3	2	1	1	1	1	1

$$\text{Numerator} = 0.96 - 0.84 + 1 = 1.12$$

$$\begin{aligned}\text{Denominator} &= \sqrt{0.36 + 1.96 + 1} \times \sqrt{2.56 + 0.36 + 1} \\ &= 1.822 \times 2.218 = 4.04\end{aligned}$$

$$\text{Sim}(P5, P4) = 1.12 / 4.04 = 0.277$$

③ Predict missing Ratings,

Prediction 1: user 2's rating for P5

user 2's rating

$$\hat{r}_{2,5} = \frac{\text{Sim}(5,1) r_{c,1} + \text{Sim}(5,2) r_{c,2} + \text{Sim}(5,3) r_{c,3} + \text{Sim}(5,4) r_{c,4}}{|\text{Sim}(2,1)| + |\text{Sim}(2,2)| + |\text{Sim}(2,3)| + |\text{Sim}(2,4)|}$$

P1	4
P2	3
P3	4
P4	1

$$= \frac{(-0.325 \times 4) + (-0.398 \times 3) + (-0.825 \times 4) + (0.277 \times 1)}{0.325 + 0.398 + 0.825 + 0.277}$$

$$= \frac{-1.30 - 1.19 - 3.30 + 0.277}{0.325 + 0.398 + 0.825 + 0.277} = \frac{-5.513}{1.825} = -3.02$$

using absolute similarity,

$$\hat{r}_{2,5} = \frac{(0.325 \times 4 + 0.398 \times 3 + 0.825 \times 4 + 0.277 \times 1)}{1.825}$$

$$= \frac{5.067}{1.825} = 3.32 \approx 3$$

$$\hat{r}_{2,5} = 3$$

Prediction 2: user 5's rating for P5

user 5's rating

$$\hat{r}_{5,5} = \frac{\text{Sim}(5,1) r_{c,1} + \text{Sim}(5,2) r_{c,2} + \text{Sim}(5,3) r_{c,3} + \text{Sim}(5,4) r_{c,4}}{|\text{Sim}(5,1)| + |\text{Sim}(5,2)| + |\text{Sim}(5,3)| + |\text{Sim}(5,4)|}$$

P1	2
P2	1
P3	5
P4	1

$$= \frac{(-0.325 \times 2) + (-0.398 \times 1) + (-0.825 \times 5) + (0.277 \times 1)}{0.325 + 0.398 + 0.825 + 0.277}$$

$$= \frac{-0.65 - 0.398 - 4.125 + 0.277}{1.825} = \frac{-4.896}{1.825} = -2.68$$

(13)

using absolute similarity,

$$\hat{\eta}_{5,5} = \frac{(0.325 \times 2) + (0.398 \times 1) + (0.825 \times 5) + (0.277 \times 1)}{1.825}$$

$$= \frac{5.45}{1.825} = 2.98 \approx 3$$

$$\hat{\eta}_{5,5} = 3$$

Final answer,

user	P1	P2	P3	P4	P5
1	3	4	3	1	2
2	4	3	4	1	3
3	3	3	1	2	4
4	1	1	2	3	3
5	2	1	5	1	3